

Received Order Analysis Engine (ROAE)

Analysis of the King Wen sequence including observations by Terence McKenna

Hexagram reference table

Hexagram reference table					
Each hexagram is a stack of 6 lines, either solid (1/yang) or broken (0/yin).					
The 'binary' column encodes these 6 lines as bits (bottom line = rightmost bit).					
Each hexagram is also composed of two 3-line trigrams: an upper and a lower.					
There are 8 possible trigrams (Heaven, Earth, Thunder, Water, Mountain, Wind, Fire, Lake), giving 8x8 = 64 possible hexagrams.					

Pos	Hex	Binary	Upper	Lower	Name
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01	䷀	111111	Qian Heaven	Qian Heaven	Heaven over Heaven
02	䷁	000000	Kun Earth	Kun Earth	Earth over Earth
03	䷂	010001	Kan Water	Zhen Thunder	Water over Thunder
04	䷃	100010	Gen Mountain	Kan Water	Mountain over Water
05	䷄	010111	Kan Water	Qian Heaven	Water over Heaven
06	䷅	111010	Qian Heaven	Kan Water	Heaven over Water
07	䷆	000010	Kun Earth	Kan Water	Earth over Water
08	䷇	010000	Kan Water	Kun Earth	Water over Earth
09	䷈	110111	Xun Wind	Qian Heaven	Wind over Heaven
10	䷉	111011	Qian Heaven	Dui Lake	Heaven over Lake
11	䷊	000111	Kun Earth	Qian Heaven	Earth over Heaven
12	䷋	111000	Qian Heaven	Kun Earth	Heaven over Earth
13	䷌	111101	Qian Heaven	Li Fire	Heaven over Fire
14	䷍	101111	Li Fire	Qian Heaven	Fire over Heaven
15	䷎	000100	Kun Earth	Gen Mountain	Earth over Mountain
16	䷏	001000	Zhen Thunder	Kun Earth	Thunder over Earth
17	䷐	011001	Dui Lake	Zhen Thunder	Lake over Thunder
18	䷑	100110	Gen Mountain	Xun Wind	Mountain over Wind
19	䷒	000011	Kun Earth	Dui Lake	Earth over Lake
20	䷓	110000	Xun Wind	Kun Earth	Wind over Earth
21	䷔	101001	Li Fire	Zhen Thunder	Fire over Thunder
22	䷕	100101	Gen Mountain	Li Fire	Mountain over Fire
23	䷖	100000	Gen Mountain	Kun Earth	Mountain over Earth
24	䷗	000001	Kun Earth	Zhen Thunder	Earth over Thunder
25	䷘	111001	Qian Heaven	Zhen Thunder	Heaven over Thunder
26	䷙	100111	Gen Mountain	Qian Heaven	Mountain over Heaven
27	䷚	100001	Gen Mountain	Zhen Thunder	Mountain over Thunder
28	䷛	011110	Dui Lake	Xun Wind	Lake over Wind
29	䷜	010010	Kan Water	Kan Water	Water over Water
30	䷝	101101	Li Fire	Li Fire	Fire over Fire
31	䷞	011100	Dui Lake	Gen Mountain	Lake over Mountain
32	䷟	001110	Zhen Thunder	Xun Wind	Thunder over Wind
33	䷠	111100	Qian Heaven	Gen Mountain	Heaven over Mountain
34	䷡	001111	Zhen Thunder	Qian Heaven	Thunder over Heaven
35	䷢	101000	Li Fire	Kun Earth	Fire over Earth
36	䷣	000101	Kun Earth	Li Fire	Earth over Fire
37	䷤	110101	Xun Wind	Li Fire	Wind over Fire
38	䷥	101011	Li Fire	Dui Lake	Fire over Lake
39	䷦	010100	Kan Water	Gen Mountain	Water over Mountain
40	䷧	001010	Zhen Thunder	Kan Water	Thunder over Water
41	䷨	100011	Gen Mountain	Dui Lake	Mountain over Lake
42	䷩	110001	Xun Wind	Zhen Thunder	Wind over Thunder
43	䷪	011111	Dui Lake	Qian Heaven	Lake over Heaven
44	䷫	111110	Qian Heaven	Xun Wind	Heaven over Wind
45	䷬	011000	Dui Lake	Kun Earth	Lake over Earth
46	䷭	000110	Kun Earth	Xun Wind	Earth over Wind
47	䷮	011010	Dui Lake	Kan Water	Lake over Water
48	䷯	010110	Kan Water	Xun Wind	Water over Wind
49	䷰	011101	Dui Lake	Li Fire	Lake over Fire
50	䷱	101110	Li Fire	Xun Wind	Fire over Wind
51	䷲	001001	Zhen Thunder	Zhen Thunder	Thunder over Thunder
52	䷳	100100	Gen Mountain	Gen Mountain	Mountain over Mountain

53				110100		Xun Wind		Gen Mountain		Wind over Mountain
54				001011		Zhen Thunder		Dui Lake		Thunder over Lake
55				001101		Zhen Thunder		Li Fire		Thunder over Fire
56				101100		Li Fire		Gen Mountain		Fire over Mountain
57				110110		Xun Wind		Xun Wind		Wind over Wind
58				011011		Dui Lake		Dui Lake		Lake over Lake
59				110010		Xun Wind		Kan Water		Wind over Water
60				010011		Kan Water		Dui Lake		Water over Lake
61				110011		Xun Wind		Dui Lake		Wind over Lake
62				001100		Zhen Thunder		Gen Mountain		Thunder over Mountain
63				010101		Kan Water		Li Fire		Water over Fire
64				101010		Li Fire		Kan Water		Fire over Water

Reverse vs. Inverse pair analysis

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Reverse vs. Inverse pair analysis
The 64 hexagrams are traditionally grouped into 32 pairs (1-2, 3-4, ..., 63-64).
For each pair, we test two relationships:
    Reverse: flip the hexagram upside down (read the lines top-to-bottom instead of
             bottom-to-top). If the second hexagram upside-down equals the first,
             they are a reverse pair.
    Inverse: toggle every line (solid becomes broken, broken becomes solid).
             If toggling all lines of one hexagram gives the other, they are inverse.
Key finding: EVERY pair in the King Wen sequence is one or the other – none are
unrelated. This perfect pairing structure is unlikely to occur by chance.
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01  Inverse 02 
03  Reverse 04 
05  Reverse 06 
07  Reverse 08 
09  Reverse 10 
11  Reverse 12 
13  Reverse 14 
15  Reverse 16 
17  Reverse 18 
19  Reverse 20 
21  Reverse 22 
23  Reverse 24 
25  Reverse 26 
27  Inverse 28 
29  Inverse 30 
31  Reverse 32 
33  Reverse 34 
35  Reverse 36 
37  Reverse 38 
39  Reverse 40 
41  Reverse 42 
43  Reverse 44 
45  Reverse 46 
47  Reverse 48 
49  Reverse 50 
51  Reverse 52 
53  Reverse 54 
55  Reverse 56 
57  Reverse 58 
59  Reverse 60 
61  Inverse 62 
63  Reverse 64 
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Reverse count pairs: 28/32 (87.5%)
Inverse count pairs: 04/32 (12.5%)
Neither count pairs: 00/32 (0.0%)
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First order of difference (the 'wave')

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First order of difference (the 'wave')
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For each consecutive pair of hexagrams, count how many of the 6 lines change.
This produces a sequence of values from 0 (identical) to 6 (every line flipped).
The resulting 'wave' is central to analysis of the King Wen sequence.
Key observation (McKenna, The Invisible Landscape, 1975): the value 5 never appears – no consecutive hexagrams differ by exactly 5 lines. The value 0 also never appears (no duplicates).

01 ䷀ to 02 ䷁ difference 6 *****
02 ䷁ to 03 ䷂ difference 2 **
03 ䷂ to 04 ䷃ difference 4 ****
04 ䷃ to 05 ䷄ difference 4 ****
05 ䷄ to 06 ䷅ difference 4 ****
06 ䷅ to 07 ䷆ difference 3 ***
07 ䷆ to 08 ䷇ difference 2 **
08 ䷇ to 09 ䷈ difference 4 ****
09 ䷈ to 10 ䷉ difference 2 **
10 ䷉ to 11 ䷊ difference 4 ****
11 ䷊ to 12 ䷋ difference 6 *****
12 ䷋ to 13 ䷌ difference 2 **
13 ䷌ to 14 ䷍ difference 2 **
14 ䷍ to 15 ䷎ difference 4 ****
15 ䷎ to 16 ䷏ difference 2 **
16 ䷏ to 17 ䷐ difference 2 **
17 ䷐ to 18 ䷑ difference 6 *****
18 ䷑ to 19 ䷒ difference 3 ***
19 ䷒ to 20 ䷓ difference 4 ****
20 ䷓ to 21 ䷔ difference 3 ***
21 ䷔ to 22 ䷕ difference 2 **
22 ䷕ to 23 ䷖ difference 2 **
23 ䷖ to 24 ䷗ difference 2 **
24 ䷗ to 25 ䷘ difference 3 ***
25 ䷘ to 26 ䷙ difference 4 ****
26 ䷙ to 27 ䷚ difference 2 **
27 ䷚ to 28 ䷛ difference 6 *****
28 ䷛ to 29 ䷜ difference 2 **
29 ䷜ to 30 ䷝ difference 6 *****
30 ䷝ to 31 ䷞ difference 3 ***
31 ䷞ to 32 ䷟ difference 2 **
32 ䷟ to 33 ䷠ difference 3 ***
33 ䷠ to 34 ䷡ difference 4 ****
34 ䷡ to 35 ䷢ difference 4 ****
35 ䷢ to 36 ䷣ difference 4 ****
36 ䷣ to 37 ䷤ difference 2 **
37 ䷤ to 38 ䷥ difference 4 ****
38 ䷥ to 39 ䷦ difference 6 *****
39 ䷦ to 40 ䷧ difference 4 ****
40 ䷧ to 41 ䷨ difference 3 ***
41 ䷨ to 42 ䷩ difference 2 **
42 ䷩ to 43 ䷪ difference 4 ****
43 ䷪ to 44 ䷫ difference 2 **
44 ䷫ to 45 ䷬ difference 3 ***
45 ䷬ to 46 ䷭ difference 4 ****
46 ䷭ to 47 ䷮ difference 3 ***
47 ䷮ to 48 ䷯ difference 2 **
48 ䷯ to 49 ䷰ difference 3 ***
49 ䷰ to 50 ䷱ difference 4 ****
50 ䷱ to 51 ䷲ difference 4 ****
51 ䷲ to 52 ䷳ difference 4 ****
52 ䷳ to 53 ䷴ difference 1 *
53 ䷴ to 54 ䷵ difference 6 *****
54 ䷵ to 55 ䷶ difference 2 **
55 ䷶ to 56 ䷷ difference 2 **
56 ䷷ to 57 ䷸ difference 3 ***
57 ䷸ to 58 ䷹ difference 4 ****
58 ䷹ to 59 ䷺ difference 3 ***
59 ䷺ to 60 ䷻ difference 2 **
60 ䷻ to 61 ䷼ difference 1 *
61 ䷼ to 62 ䷽ difference 6 *****
62 ䷽ to 63 ䷾ difference 3 ***
63 ䷾ to 64 ䷿ difference 6 *****

0 line changes total 0
1 line changes total 2
2 line changes total 20
3 line changes total 13
4 line changes total 19

6 line changes total 9

Age Group	Percentage
18-24	10
25-34	25
35-44	25
45-54	35
55-64	20
65-74	10
75-84	10
85-94	10
95+	10

Note: does not survive Bonferroni correction for 28 tests (threshold $p < 0.0018$).

	Kun	Zhen	Kan	Dui	Gen	Li	Xun	Qian
Kun	0	1	2	1	0	0	2	2
Zhen	0	1	1	1	2	2	0	1
Kan	0	1	0	1	1	2	2	1
Dui	1	1	2	0	1	1	1	1
Gen	2	0	1	1	2	0	2	0
Li	2	1	1	1	1	0	1	0
Xun	0	2	1	2	0	2	0	1
Qian	3	1	0	1	1	1	0	1

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--- Lower trigram transition matrix ---
Rows = from, Columns = to
      Kun   Zhen Kan   Dui   Gen   Li   Xun   Qian
Kun    0     4     0     0     0     2     1     1
Zhen    0     1     1     0     1     1     2     2
Kan     1     0     1     2     0     1     1     1
Dui     1     1     1     1     2     1     0     1
Gen     1     0     1     1     1     1     2     1
Li      1     0     1     1     2     1     1     1
Xun     1     1     2     2     1     1     0     0
Qian    3     1     1     1     1     0     1     0

--- Trigram change rates vs null (pair-preserving permutation test) ---
Null: 10,000 orderings preserving the pair structure (permute pairs + flip orientations),
the correct baseline per CRITIQUE.md. Statistic: upper/lower trigram change counts.
KW upper changes 59/63: percentile 47.3 (fraction of null <= KW)
KW lower changes 58/63: percentile 27.3

--- Pure (doubled-trigram) hexagram placement ---
The 8 pure hexagrams (upper == lower trigram). Lai Zhide (1525-1604, via Schulz 1982)
observed kan/li doubles closing both Classics; measured here against the same null.
Pure hexagram positions (1-based): [1, 2, 29, 30, 51, 52, 57, 58]
Pure hexagrams at Classic ends (positions 1,2,29,30,63,64): 4 of a possible 6
Null P(>= KW's 4) = 0.0336 (pair-preserving null; note KW's C4 fixes 1,2 by
definition, so interpret against the constrained baseline)

--- Nuclear trigram structure ---
Nuclear hexagram = lines 2-4 (lower nuclear) + 3-5 (upper nuclear). Classical fact
(commentary tradition): iterating the nuclear map sends all 64 hexagrams to 16, then
to a 4-element terminal attractor {0, 21, 42, 63}. CORRECTED 2026-08-01: an earlier
version called these '4 fixed points', which is false and self-contradicting -- it
then called two of them 'alternators'. There are exactly TWO fixed points, 0 and 63;
21 and 42 form a 2-cycle (nuclear(21) = 42, nuclear(42) = 21). The 16/4/4 counts below
are unaffected.
Distinct after 1 application: 16; after 2: 4; after 3: 4 -> [0, 21, 42, 63]
KW nuclear-hexagram changes along the sequence: 58/63

--- Jing Fang Eight-Palaces rank correlation ---
Spearman rank correlation between KW positions and Jing Fang's palace ordering
(the trigram-generated classical alternative; ordering as in solve.c --null-historical).
Spearman rho = 0.1384; null P(|rho| >= observed) = 0.1170 (pair-preserving null)

--- Symmetry group vs the trigram decomposition (ROAE SYMMETRY_SEARCH theorem) ---
The C1-C5 symmetry group B3 (order 48) acts on LINE positions; reversal maps the upper
trigram to the reversed lower trigram, so most of the group does NOT preserve the
upper/lower split. Preserving elements = permutations acting within {0,1,2} and {3,4,5}
blocks (or swapping them wholesale) that also centralize rev.
Split-respecting subgroup order: 12 of 48 (computed over the rev-centralizer)

--- Methodological note ---
With ~1 expected observation per cell, no goodness-of-fit test (e.g.,
chi-square) has sufficient power to detect deviations from uniform
transitions. The matrices are descriptive only.
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Nuclear hexagram analysis

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Nuclear hexagram analysis
Every hexagram contains a hidden 'nuclear' hexagram inside it, formed by
extracting the 4 middle lines (lines 2-3-4-5). Lines 2-3-4 form the lower
nuclear trigram, and lines 3-4-5 form the upper nuclear trigram (sharing lines
3 and 4). This creates a chain: each hexagram points to its nuclear hexagram,
which points to its own nuclear hexagram, and so on until the chain loops.
These chains reveal deep structural relationships between hexagrams that are
not obvious from the surface arrangement.
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01 ䷀ nuclear: 01 ䷀ chain: 01 ䷀ -> 01 ䷀
02 ䷁ nuclear: 02 ䷁ chain: 02 ䷁ -> 02 ䷁
03 ䷂ nuclear: 23 ䷂ chain: 03 ䷂ -> 23 ䷂ -> 02 ䷁ -> 02 ䷁
04 ䷃ nuclear: 24 ䷃ chain: 04 ䷃ -> 24 ䷃ -> 02 ䷁ -> 02 ䷁
05 ䷄ nuclear: 38 ䷄ chain: 05 ䷄ -> 38 ䷄ -> 63 ䷮ -> 64 ䷯ -> 63 ䷮
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06	nuclear: 37	chain: 06	-> 37	-> 64	-> 63	-> 64
07	nuclear: 24	chain: 07	-> 24	-> 02	-> 02	
08	nuclear: 23	chain: 08	-> 23	-> 02	-> 02	
09	nuclear: 38	chain: 09	-> 38	-> 63	-> 64	-> 63
10	nuclear: 37	chain: 10	-> 37	-> 64	-> 63	-> 64
11	nuclear: 54	chain: 11	-> 54	-> 63	-> 64	-> 63
12	nuclear: 53	chain: 12	-> 53	-> 64	-> 63	-> 64
13	nuclear: 44	chain: 13	-> 44	-> 01	-> 01	
14	nuclear: 43	chain: 14	-> 43	-> 01	-> 01	
15	nuclear: 40	chain: 15	-> 40	-> 63	-> 64	-> 63
16	nuclear: 39	chain: 16	-> 39	-> 64	-> 63	-> 64
17	nuclear: 53	chain: 17	-> 53	-> 64	-> 63	-> 64
18	nuclear: 54	chain: 18	-> 54	-> 63	-> 64	-> 63
19	nuclear: 24	chain: 19	-> 24	-> 02	-> 02	
20	nuclear: 23	chain: 20	-> 23	-> 02	-> 02	
21	nuclear: 39	chain: 21	-> 39	-> 64	-> 63	-> 64
22	nuclear: 40	chain: 22	-> 40	-> 63	-> 64	-> 63
23	nuclear: 02	chain: 23	-> 02	-> 02		
24	nuclear: 02	chain: 24	-> 02	-> 02		
25	nuclear: 53	chain: 25	-> 53	-> 64	-> 63	-> 64
26	nuclear: 54	chain: 26	-> 54	-> 63	-> 64	-> 63
27	nuclear: 02	chain: 27	-> 02	-> 02		
28	nuclear: 01	chain: 28	-> 01	-> 01		
29	nuclear: 27	chain: 29	-> 27	-> 02	-> 02	
30	nuclear: 28	chain: 30	-> 28	-> 01	-> 01	
31	nuclear: 44	chain: 31	-> 44	-> 01	-> 01	
32	nuclear: 43	chain: 32	-> 43	-> 01	-> 01	
33	nuclear: 44	chain: 33	-> 44	-> 01	-> 01	
34	nuclear: 43	chain: 34	-> 43	-> 01	-> 01	
35	nuclear: 39	chain: 35	-> 39	-> 64	-> 63	-> 64
36	nuclear: 40	chain: 36	-> 40	-> 63	-> 64	-> 63
37	nuclear: 64	chain: 37	-> 64	-> 63	-> 64	
38	nuclear: 63	chain: 38	-> 63	-> 64	-> 63	
39	nuclear: 64	chain: 39	-> 64	-> 63	-> 64	
40	nuclear: 63	chain: 40	-> 63	-> 64	-> 63	
41	nuclear: 24	chain: 41	-> 24	-> 02	-> 02	
42	nuclear: 23	chain: 42	-> 23	-> 02	-> 02	
43	nuclear: 01	chain: 43	-> 01	-> 01		
44	nuclear: 01	chain: 44	-> 01	-> 01		
45	nuclear: 53	chain: 45	-> 53	-> 64	-> 63	-> 64
46	nuclear: 54	chain: 46	-> 54	-> 63	-> 64	-> 63
47	nuclear: 37	chain: 47	-> 37	-> 64	-> 63	-> 64
48	nuclear: 38	chain: 48	-> 38	-> 63	-> 64	-> 63
49	nuclear: 44	chain: 49	-> 44	-> 01	-> 01	
50	nuclear: 43	chain: 50	-> 43	-> 01	-> 01	
51	nuclear: 39	chain: 51	-> 39	-> 64	-> 63	-> 64
52	nuclear: 40	chain: 52	-> 40	-> 63	-> 64	-> 63
53	nuclear: 64	chain: 53	-> 64	-> 63	-> 64	
54	nuclear: 63	chain: 54	-> 63	-> 64	-> 63	
55	nuclear: 28	chain: 55	-> 28	-> 01	-> 01	
56	nuclear: 28	chain: 56	-> 28	-> 01	-> 01	
57	nuclear: 38	chain: 57	-> 38	-> 63	-> 64	-> 63
58	nuclear: 37	chain: 58	-> 37	-> 64	-> 63	-> 64
59	nuclear: 27	chain: 59	-> 27	-> 02	-> 02	
60	nuclear: 27	chain: 60	-> 27	-> 02	-> 02	
61	nuclear: 27	chain: 61	-> 27	-> 02	-> 02	
62	nuclear: 28	chain: 62	-> 28	-> 01	-> 01	
63	nuclear: 64	chain: 63	-> 64	-> 63		
64	nuclear: 63	chain: 64	-> 63	-> 64		

--- Nuclear hexagram frequency ---

How often each hexagram appears as another's nuclear hexagram

01	Heaven over Heaven	appears 4 times
02	Earth over Earth	appears 4 times
23	Mountain over Earth	appears 4 times
24	Earth over Thunder	appears 4 times
27	Mountain over Thunder	appears 4 times
28	Lake over Wind	appears 4 times
37	Wind over Fire	appears 4 times
38	Fire over Lake	appears 4 times
39	Water over Mountain	appears 4 times
40	Thunder over Water	appears 4 times
43	Lake over Heaven	appears 4 times
44	Heaven over Wind	appears 4 times
53	Wind over Mountain	appears 4 times

54 ䷒ Thunder over Lake appears 4 times

63 ䷒ Water over Fire appears 4 times

64 ䷒ Fire over Water appears 4 times

--- Null model note ---

Nuclear hexagram derivation is a fixed function of binary value (lines 2-5), independent of the King Wen ordering. The chain structure, cycle lengths, and frequency distribution above are identical for ANY ordering of the 64 hexagrams. These properties reflect the 6-bit binary system, not the King Wen sequence specifically.

Line change positional analysis

Line change positional analysis

Each hexagram has 6 lines (positions 1-6, bottom to top). When consecutive hexagrams differ, which specific lines are changing? If the King Wen sequence were random, each line position would change about 50% of the time. Significant deviation from 50% would suggest the ordering favors changing certain positions. The co-change matrix shows which pairs of lines tend to flip together in the same transition – revealing correlated line movements.

Line 1: 37/63 (58.7%) #####

Line 2: 32/63 (50.8%) #####

Line 3: 35/63 (55.6%) #####

Line 4: 34/63 (54.0%) #####

Line 5: 35/63 (55.6%) #####

Line 6: 38/63 (60.3%) #####

--- Line co-change frequency ---

How often pairs of lines change together in the same transition

	L1	L2	L3	L4	L5	L6
L1	.	18	20	19	19	30
L2	.	.	20	17	23	20
L3	.	.	.	25	16	21
L4	.	.	.	.	20	20
L5	.	.	.	.	.	20
L6	.	.	.	.	.	.

Complement distance analysis

Complement distance analysis

Every hexagram has an 'opposite' (complement) formed by toggling all 6 lines (solid becomes broken, broken becomes solid). For example, hexagram 1 (all solid) and hexagram 2 (all broken) are complements. This section finds where each hexagram's complement sits in the King Wen sequence and how far apart they are. If complements sit nearer each other than a null model predicts, opposition is an organizing feature of the sequence's structure; if farther, complements may serve as bookends for larger structural sections.

01 ䷀ <-> 02 ䷁ distance: 1 Heaven over Heaven <-> Earth over Earth

02 ䷁ <-> 01 ䷀ distance: 1 Earth over Earth <-> Heaven over Heaven

03 ䷓ <-> 50 ䷋ distance: 47 Water over Thunder <-> Fire over Wind

04 ䷓ <-> 49 ䷊ distance: 45 Mountain over Water <-> Lake over Fire

05 ䷒ <-> 35 ䷖ distance: 30 Water over Heaven <-> Fire over Earth

06 ䷒ <-> 36 ䷔ distance: 30 Heaven over Water <-> Earth over Fire

07 ䷒ <-> 13 ䷗ distance: 6 Earth over Water <-> Heaven over Fire

08 ䷒ <-> 14 ䷔ distance: 6 Water over Earth <-> Fire over Heaven

09 ䷒ <-> 16 ䷄ distance: 7 Wind over Heaven <-> Thunder over Earth

10 ䷒ <-> 15 ䷄ distance: 5 Heaven over Lake <-> Earth over Mountain

11 ䷒ <-> 12 ䷒ distance: 1 Earth over Heaven <-> Heaven over Earth

12 ䷒ <-> 11 ䷒ distance: 1 Heaven over Earth <-> Earth over Heaven

13 ䷒ <-> 07 ䷒ distance: 6 Heaven over Fire <-> Earth over Water

14 ䷒ <-> 08 ䷒ distance: 6 Fire over Heaven <-> Water over Earth

15 ䷒ <-> 10 ䷒ distance: 5 Earth over Mountain <-> Heaven over Lake

16 ䷒ <-> 09 ䷒ distance: 7 Thunder over Earth <-> Wind over Heaven

17 ䷒ <-> 18 ䷔ distance: 1 Lake over Thunder <-> Mountain over Wind

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18 ䷮ <-> 17 ䷮ distance: 1 Mountain over Wind <-> Lake over Thunder
19 ䷮ <-> 33 ䷮ distance: 14 Earth over Lake <-> Heaven over Mountain
20 ䷮ <-> 34 ䷮ distance: 14 Wind over Earth <-> Thunder over Heaven
21 ䷮ <-> 48 ䷮ distance: 27 Fire over Thunder <-> Water over Wind
22 ䷮ <-> 47 ䷮ distance: 25 Mountain over Fire <-> Lake over Water
23 ䷮ <-> 43 ䷮ distance: 20 Mountain over Earth <-> Lake over Heaven
24 ䷮ <-> 44 ䷮ distance: 20 Earth over Thunder <-> Heaven over Wind
25 ䷮ <-> 46 ䷮ distance: 21 Heaven over Thunder <-> Earth over Wind
26 ䷮ <-> 45 ䷮ distance: 19 Mountain over Heaven <-> Lake over Earth
27 ䷮ <-> 28 ䷮ distance: 1 Mountain over Thunder <-> Lake over Wind
28 ䷮ <-> 27 ䷮ distance: 1 Lake over Wind <-> Mountain over Thunder
29 ䷮ <-> 30 ䷮ distance: 1 Water over Water <-> Fire over Fire
30 ䷮ <-> 29 ䷮ distance: 1 Fire over Fire <-> Water over Water
31 ䷮ <-> 41 ䷮ distance: 10 Lake over Mountain <-> Mountain over Lake
32 ䷮ <-> 42 ䷮ distance: 10 Thunder over Wind <-> Wind over Thunder
33 ䷮ <-> 19 ䷮ distance: 14 Heaven over Mountain <-> Earth over Lake
34 ䷮ <-> 20 ䷮ distance: 14 Thunder over Heaven <-> Wind over Earth
35 ䷮ <-> 05 ䷮ distance: 30 Fire over Earth <-> Water over Heaven
36 ䷮ <-> 06 ䷮ distance: 30 Earth over Fire <-> Heaven over Water
37 ䷮ <-> 40 ䷮ distance: 3 Wind over Fire <-> Thunder over Water
38 ䷮ <-> 39 ䷮ distance: 1 Fire over Lake <-> Water over Mountain
39 ䷮ <-> 38 ䷮ distance: 1 Water over Mountain <-> Fire over Lake
40 ䷮ <-> 37 ䷮ distance: 3 Thunder over Water <-> Wind over Fire
41 ䷮ <-> 31 ䷮ distance: 10 Mountain over Lake <-> Lake over Mountain
42 ䷮ <-> 32 ䷮ distance: 10 Wind over Thunder <-> Thunder over Wind
43 ䷮ <-> 23 ䷮ distance: 20 Lake over Heaven <-> Mountain over Earth
44 ䷮ <-> 24 ䷮ distance: 20 Heaven over Wind <-> Earth over Thunder
45 ䷮ <-> 26 ䷮ distance: 19 Lake over Earth <-> Mountain over Heaven
46 ䷮ <-> 25 ䷮ distance: 21 Earth over Wind <-> Heaven over Thunder
47 ䷮ <-> 22 ䷮ distance: 25 Lake over Water <-> Mountain over Fire
48 ䷮ <-> 21 ䷮ distance: 27 Water over Wind <-> Fire over Thunder
49 ䷮ <-> 04 ䷮ distance: 45 Lake over Fire <-> Mountain over Water
50 ䷮ <-> 03 ䷮ distance: 47 Fire over Wind <-> Water over Thunder
51 ䷮ <-> 57 ䷮ distance: 6 Thunder over Thunder <-> Wind over Wind
52 ䷮ <-> 58 ䷮ distance: 6 Mountain over Mountain <-> Lake over Lake
53 ䷮ <-> 54 ䷮ distance: 1 Wind over Mountain <-> Thunder over Lake
54 ䷮ <-> 53 ䷮ distance: 1 Thunder over Lake <-> Wind over Mountain
55 ䷮ <-> 59 ䷮ distance: 4 Thunder over Fire <-> Wind over Water
56 ䷮ <-> 60 ䷮ distance: 4 Fire over Mountain <-> Water over Lake
57 ䷮ <-> 51 ䷮ distance: 6 Wind over Wind <-> Thunder over Thunder
58 ䷮ <-> 52 ䷮ distance: 6 Lake over Lake <-> Mountain over Mountain
59 ䷮ <-> 55 ䷮ distance: 4 Wind over Water <-> Thunder over Fire
60 ䷮ <-> 56 ䷮ distance: 4 Water over Lake <-> Fire over Mountain
61 ䷮ <-> 62 ䷮ distance: 1 Wind over Lake <-> Thunder over Mountain
62 ䷮ <-> 61 ䷮ distance: 1 Thunder over Mountain <-> Wind over Lake
63 ䷮ <-> 64 ䷮ distance: 1 Water over Fire <-> Fire over Water
64 ䷮ <-> 63 ䷮ distance: 1 Fire over Water <-> Water over Fire
```

--- Complement distance statistics ---

Mean distance: 12.1

Median distance: 6

Min distance: 1 (adjacent = pair is an inverse pair)

Max distance: 47

--- Complement distance null model ---

Random mean complement distance (over 10000 shuffles): 21.7

King Wen mean complement distance: 12.1

King Wen percentile vs random: 0.0%

Complements are significantly closer together than chance would predict:
the ordering keeps complements unusually near one another.

Palindrome analysis of the difference wave

Palindrome analysis of the difference wave

A palindrome reads the same forwards and backwards (like 2,4,6,4,2). Palindromic runs in the difference wave are mirror symmetry – sections where the pattern of change rises and falls in a balanced way. Longer palindromes are less likely to occur by chance, so counts and lengths are measured against null models below.

Found 27 palindromic subsequences (showing longest 20)

Length 7 at positions 11-17: [6,2,2,4,2,2,6] 

Length 7 at positions 19-25: [4,3,2,2,2,3,4] 

Length 7 at positions 39-45: [4,3,2,4,2,3,4] 

Length 5 at positions 12-16: [2,2,4,2,2] 

Length 5 at positions 20-24: [3,2,2,2,3] 

Length 5 at positions 40-44: [3,2,4,2,3] 

Length 5 at positions 43-47: [2,3,4,3,2] 

Length 5 at positions 45-49: [4,3,2,3,4] 

Length 5 at positions 55-59: [2,3,4,3,2] 

Length 3 at positions 03-05: [4,4,4] 

Length 3 at positions 07-09: [2,4,2] 

Length 3 at positions 08-10: [4,2,4] 

Length 3 at positions 13-15: [2,4,2] 

Length 3 at positions 18-20: [3,4,3] 

Length 3 at positions 21-23: [2,2,2] 

Length 3 at positions 26-28: [2,6,2] 

Length 3 at positions 27-29: [6,2,6] 

Length 3 at positions 30-32: [3,2,3] 

Length 3 at positions 33-35: [4,4,4] 

Length 3 at positions 35-37: [4,2,4] 

--- Palindrome count by length ---
Length 7: 3 found
Length 5: 6 found
Length 3: 18 found

--- Palindrome null model ---
How many palindromes of length >= 3 do random permutations produce?
King Wen palindromes: 27, longest: 7
Random mean palindromes: 22.0, mean longest: 6.8
King Wen palindrome count percentile: 80.1%
King Wen longest palindrome percentile: 44.9%

--- Pair-constrained palindrome null model ---
The unconstrained comparison shuffles all 64 hexagrams freely. A fairer comparison preserves the pair structure: each pair of hexagrams stays adjacent, only the pair order and within-pair orientation are randomized.
King Wen palindromes: 27, longest: 7
Pair-constrained mean palindromes: 27.7, mean longest: 8.5
King Wen palindrome count percentile (pair-constrained): 48.6%
King Wen longest palindrome percentile (pair-constrained): 13.1%

Upper Canon (1-30) vs. Lower Canon (31-64) comparison

Upper Canon (1-30) vs. Lower Canon (31-64) comparison
The I Ching is traditionally divided into two books: the Upper Canon (hexagrams 1-30) and Lower Canon (hexagrams 31-64). This division is ancient and may reflect different authorship, time periods, or thematic groupings. This section compares the mathematical properties of each half: do they have similar difference wave patterns? Similar yin/yang balance? Does the 'no 5-line transition' property hold in each half independently, or only in the whole?

Metric	Upper (1-30)	Lower (31-64)
Hexagrams	30	34
Transitions	29	33
Mean line changes	3.38	3.33
Yang lines (total)	86	106
Yin lines (total)	94	98
Difference distribution:	Upper (1-30)	Lower (31-64)
0 line changes	0	0
1 line changes	0	2
2 line changes	12	8
3 line changes	4	8
4 line changes	8	11
5 line changes	0	0
6 line changes	5	4



5-line transitions in Upper Canon: 0
5-line transitions in Lower Canon: 0

--- Canon split null model ---
Permutation test: is the King Wen split at position 30 special, or would any random split of the 64-hexagram sequence show a similar gap in mean line-change differences between the two halves?
King Wen |upper_mean - lower_mean|: 0.0460
Random permutations with gap >= King Wen: 8755/10000
King Wen gap percentile: 12.5%
The King Wen canon split does not produce a statistically significant mean-difference gap. The observed difference between upper and lower halves is consistent with what random shuffles produce.

Autocorrelation of the difference wave

Autocorrelation of the difference wave
Autocorrelation measures whether the wave is correlated with a shifted copy of itself. A peak at lag N means the pattern tends to repeat every N steps. A value near +1 means strong repetition, near -1 means strong alternation, and near 0 means no relationship. If the King Wen ordering contains hidden periodic structure, it will show up as peaks in this analysis.
Caveat: with only N=63 data points, the 95% noise threshold is +/-0.25. Values within that band are indistinguishable from random noise.

Mean difference: 3.349
Variance: 1.910
95% noise threshold: +/-0.247 (values inside this are not significant)

Lag	Autocorrelation	Sig?	Visualization
---	-----	----	-----
0	1.0000	[	: : #]
1	-0.2469	[	# :]
2	-0.0098	[	: :]
3	-0.0491	[	: :]
4	-0.0447	[	: :]
5	-0.0457	[	: :]
6	-0.0218	[	: :]
7	-0.0644	[	: :]
8	0.1204	[	: # :]
9	-0.1275	[	: # :]
10	0.1375	[	: #:]
11	-0.0290	[	: :]
12	-0.0221	[	: :]
13	0.0850	[	: # :]
14	-0.1596	[	:# :]
15	-0.0110	[	: :]
16	0.1015	[	: # :]
17	-0.1514	[	:# :]
18	0.0055	[	: :]
19	0.0851	[	: # :]
20	-0.1403	[	:# :]
21	0.0274	[	: :]
22	-0.1315	[	: # :]
23	-0.0137	[	: :]
24	0.2263	[	: #]
25	-0.1595	[	:# :]
26	0.1112	[	: # :]
27	-0.1109	[	: # :]
28	0.1649	[	: #:]
29	0.0093	[	: :]
30	-0.0969	[	: # :]
31	-0.0480	[	: :]

Significant lags (outside noise band): 0/31
(Expected false positives at 95% threshold with 31 lags: ~1.6)
No significant autocorrelation detected – consistent with no hidden periodicity,

but note that N=63 provides limited statistical power to detect weak periodicity.

Spectral analysis (Discrete Fourier Transform)

Spectral analysis (Discrete Fourier Transform)
The DFT decomposes the difference wave into frequency components, like splitting white light into a rainbow. Each frequency represents a periodic pattern in the wave. A strong peak at frequency F means the wave has a repeating cycle every 63/F transitions. If the King Wen ordering has hidden periodic structure, the spectrum will show clear peaks rather than flat noise.
Caveat: with only 63 samples, each of the 31 frequency bins has wide uncertainty. A flat spectrum may indicate no periodicity, or insufficient data.

Number of samples: 63
Mean (DC component): 3.349
White noise floor: 0.1741 (magnitudes near this are not significant)

Freq	Period	Magnitude	Sig?	Spectrum
----	-----	-----	----	-----
1	63.0	0.0224		##
2	31.5	0.1823		#####
3	21.0	0.0145		#
4	15.8	0.1359		#####
5	12.6	0.2021		#####
6	10.5	0.1157		#####
7	9.0	0.2399		#####
8	7.9	0.1092		#####
9	7.0	0.0553		####
10	6.3	0.1630		#####
11	5.7	0.2371		#####
12	5.2	0.0473		####
13	4.8	0.2326		#####
14	4.5	0.0331		###
15	4.2	0.1145		#####
16	3.9	0.2204		#####
17	3.7	0.1451		#####
18	3.5	0.1975		#####
19	3.3	0.2197		#####
20	3.1	0.2166		#####
21	3.0	0.1611		#####
22	2.9	0.0222		##
23	2.7	0.1210		#####
24	2.6	0.4266	*	#####
25	2.5	0.0306		##
26	2.4	0.2297		#####
27	2.3	0.0795		#####
28	2.2	0.0639		####
29	2.2	0.2114		#####
30	2.1	0.2129		#####
31	2.0	0.2033		#####

Frequencies above 2x noise floor: 1/31
(The 2x threshold is ad hoc. A proper test would use Fisher's g-statistic or Bonferroni correction across frequency bins. With only 63 samples, even real periodicity may not rise above the noise floor.)

Markov chain analysis

Markov chain analysis
If we treat each difference value as a 'state', we can ask: does the current difference predict the next one? In a random sequence, knowing that the last transition changed 2 lines tells you nothing about the next transition. But if certain patterns like '6 is always followed by 2' emerge, it reveals hidden rules in the King Wen ordering. The matrix shows the probability of each transition between difference values.

```

Transition probability matrix (row = from, column = to)
      -> 1      -> 2      -> 3      -> 4      -> 5      -> 6
1      0.00      0.00      0.00      0.00      1.00      (n=2)
2      0.05      0.25      0.25      0.30      0.15      (n=20)
3      0.00      0.46      0.00      0.46      0.08      (n=13)
4      0.05      0.26      0.26      0.32      0.11      (n=19)
6      0.00      0.50      0.38      0.12      0.00      (n=8)

--- Notable transition patterns ---
(Caveat: rows with small n are unreliable – patterns based on <5 observations
are anecdotes, not statistically meaningful.)
  3 -> 2: 46% (6/13 times)
  3 -> 4: 46% (6/13 times)

--- Permutation test ---
Is the transition matrix more concentrated than random orderings produce?
King Wen matrix concentration: 2.3303
Random orderings equally or more concentrated: 2160/5000 (43.2%)

```

```

---
Gray code comparison
A Gray code is an ordering of binary values where each consecutive pair differs
by exactly 1 bit. For 6-bit hexagrams, a Gray code path would change exactly 1
line at every step – the smoothest possible sequence. The King Wen sequence does
NOT follow a Gray code (most transitions change 2-6 lines). This comparison
shows how much 'rougher' King Wen is than the theoretical minimum, suggesting
the ordering prioritizes something other than smooth transitions.
---
Metric                                Gray Code    King Wen
-----
Total path length                      63           211
Mean change per transition             1.00         3.35
Max single transition                   1            6
Min single transition                   1            1
King Wen / Gray Code ratio              3.35x

Difference distribution:                Gray Code    King Wen
1 line changes                         63           2
2 line changes                         0           20
3 line changes                         0           13
4 line changes                         0           19
6 line changes                         0            9

Gray code wave:
King Wen wave:

```

```

---
Symmetry group analysis (XOR algebra)
The 64 hexagrams form a mathematical group under XOR: combining any two hexagrams
by toggling their differing lines always produces another valid hexagram. This is
like mixing colors – every combination is itself a color. Under this algebra,
certain sets of hexagrams form 'subgroups' (closed sets where combining any two
members stays in the set). This section checks whether the King Wen pairs, inverse
pairs, and the canonical divisions correspond to meaningful algebraic subgroups.
---
XOR identity element: 0b000000 (hexagram 2, all broken)

--- Fixed points under complement (XOR 0b111111) ---
Hexagrams that are their own complement do not exist (since XOR with

```

0b111111 always flips all bits, no 6-bit value equals its own complement).

--- King Wen pair XOR products ---

For each pair, XOR the two hexagrams to see what 'difference element' they produce.
If many pairs produce the same XOR product, the pairing has algebraic regularity.

XOR Product	Hexagram	Pair count	Pairs
001100	62 ䷛	4	9-10, 15-16, 21-22, 47-48
010010	29 ䷚	4	7-8, 13-14, 31-32, 41-42
011110	28 ䷛	4	25-26, 37-38, 39-40, 45-46
100001	27 ䷚	4	23-24, 43-44, 55-56, 59-60
101101	30 ䷚	4	5-6, 35-36, 51-52, 57-58
110011	61 ䷚	4	3-4, 19-20, 33-34, 49-50
111111	01 ䷁	8	1-2, 11-12, 17-18, 27-28, 29-30, 53-54, 61-62, 63-64

Unique XOR products: 7 (out of 32 pairs)
If this number is small, the pairs share algebraic structure.

--- Inverse pair XOR closure test ---

Inverse pair hexagrams (16 total): 01, 02, 11, 12, 17, 18, 27, 28, 29, 30, 53, 54, 61, 62, 63, 64
Closed under XOR: Yes – forms a subgroup

Alternative sequence comparison

Alternative sequence comparison

The King Wen ordering is not the only way to arrange 64 hexagrams. The Fu Xi (binary) sequence orders them by numerical value (0-63), which is mathematically natural but has no traditional significance. The Mawangdui sequence was found on silk manuscripts in a 168 BCE tomb and may represent an independent tradition. Comparing the same analyses across orderings reveals what is unique to King Wen.

Metric	King Wen	Fu Xi (binary)	Mawangdui
Total path length	211	120	141
Mean change	3.35	1.90	2.24
1-line transitions	2	32	21
2-line transitions	20	16	10
3-line transitions	13	8	29
4-line transitions	19	4	2
5-line transitions	0	2	1
6-line transitions	9	1	0

Waves:



--- What makes King Wen unique ---

Zero 5-line transitions: King Wen=0, Fu Xi=2, Mawangdui=1
Zero 0-line transitions: King Wen=0, Fu Xi=0, Mawangdui=0

Windowed entropy analysis

Windowed entropy analysis

Instead of one entropy value for the whole wave, we slide a window across it and compute entropy locally. This reveals WHERE in the sequence structure concentrates (low entropy = predictable) vs. dissolves (high entropy = varied). Dips in the curve mark regions with repetitive transition patterns.

Window size: 15		
Center	Entropy	Visualization
-----	-----	-----
8	1.7056	#####
9	1.5628	#####
10	1.7056	#####
11	1.8323	#####

12	1.8323	#####
13	1.8892	#####
14	1.7968	#####
15	1.7968	#####
16	1.7232	#####
17	1.8295	#####
18	1.8295	#####
19	1.6729	#####
20	1.8295	#####
21	1.8295	#####
22	1.8295	#####
23	1.8892	#####
24	1.8892	#####
25	1.8323	#####
26	1.8892	#####
27	1.8892	#####
28	1.8892	#####
29	1.8892	#####
30	1.9086	#####
31	1.9656	#####
32	1.8892	#####
33	1.9656	#####
34	1.9656	#####
35	1.8892	#####
36	1.8892	#####
37	1.8062	#####
38	1.7465	#####
39	1.7465	#####
40	1.7465	#####
41	1.8062	#####
42	1.8062	#####
43	1.8062	#####
44	1.7465	#####
45	2.0226	#####
46	2.0226	#####
47	2.0662	#####
48	2.0419	#####
49	2.0662	#####
50	2.0662	#####
51	2.0419	#####
52	2.0662	#####
53	2.1736	#####
54	2.2566	#####
55	2.2566	#####
56	2.2892	#####

Mean windowed entropy: 1.8956
Min entropy: 1.5628 at position 9 (most structured)
Max entropy: 2.2892 at position 56 (most varied)

Entropy spark: 

--- Methodological note ---
Windowed entropy is an exploratory visualization, not a statistical test.
Apparent patterns (dips and peaks) are expected from random variation in a short sequence. Without a null model, no significance can be assigned to specific regions. Interpret as descriptive, not inferential.

Mutual information: upper vs. lower trigram transitions

Mutual information: upper vs. lower trigram transitions
When two consecutive hexagrams differ, do the upper and lower trigrams change independently, or is knowing that the upper trigram changed informative about whether the lower trigram also changed? Mutual information quantifies this:
 0 bits = completely independent (knowing one tells you nothing about the other)
 higher = correlated (they tend to change together or stay together)

Joint distribution of trigram changes:

	Lower unchanged	Lower changed
Upper unchanged	0 (0.0%)	4 (6.3%)

Upper changed5 (7.9%)54 (85.7%)

Mutual information: 0.0078 bits
Normalized MI: 0.0230 (0=independent, 1=perfectly correlated)

Mean MI of random permutations: 0.0202 bits
King Wen percentile: 7.7%

--- Full 8-state trigram mutual information ---
Using actual trigram identities (8 possible values each) rather than binary changed/unchanged, which preserves more information.
MI(upper, lower) across all 64 hexagrams: 0.000000 bits
(Expected: 0.0 – since all 64 combinations appear exactly once, the trigrams are perfectly independent in the static set.)
Note: with all 64 upper/lower trigram combinations present exactly once, independence is expected by construction (complete Latin square).

Yin-yang balance wave

Yin-yang balance wave
Each hexagram has 6 lines that are either yang (solid, 1) or yin (broken, 0).
Note: since the King Wen sequence contains all 64 possible hexagrams exactly once, the total yin-yang count is necessarily balanced (192 each). This is a mathematical tautology, not a property of the ordering. However, the LOCAL balance as you move through the sequence IS a property of the ordering – it shows where yang-dominant or yin-dominant hexagrams cluster together.

Total yang lines: 192 / 384 (50.0%)
Total yin lines: 192 / 384 (50.0%)
Mean yang per hexagram: 3.00

Running yang average (window=8):

Pos	Yang	RunAvg	Visualization
1	6	6.00	[#]
2	0	3.00	[#]
3	2	2.67	[#]
4	2	2.50	[#]
5	4	2.80	[#]
6	4	3.00	[#]
7	1	2.71	[#]
8	1	2.50	[#]
9	5	2.38	[#]
10	5	3.00	[#]
11	3	3.12	[#]
12	3	3.25	[#]
13	5	3.38	[#]
14	5	3.50	[#]
15	1	3.50	[#]
16	1	3.50	[#]
17	3	3.25	[#]
18	3	3.00	[#]
19	2	2.88	[#]
20	2	2.75	[#]
21	3	2.50	[#]
22	3	2.25	[#]
23	1	2.25	[#]
24	1	2.25	[#]
25	4	2.38	[#]
26	4	2.50	[#]
27	2	2.50	[#]
28	4	2.75	[#]
29	2	2.62	[#]
30	4	2.75	[#]
31	3	3.00	[#]
32	3	3.25	[#]
33	4	3.25	[#]
34	4	3.25	[#]
35	2	3.25	[#]
36	2	3.00	[#]
37	4	3.25	[#]

```
38      4      3.25 [          |#          ]
39      2      3.12 [          #          ]
40      2      3.00 [          #          ]
41      3      2.88 [          #|         ]
42      3      2.75 [          # |        ]
43      5      3.12 [          #          ]
44      5      3.50 [          | #         ]
45      2      3.25 [          |#         ]
46      2      3.00 [          #          ]
47      3      3.12 [          #          ]
48      3      3.25 [          |#         ]
49      4      3.38 [          | #         ]
50      4      3.50 [          | #         ]
51      2      3.12 [          #          ]
52      2      2.75 [          # |        ]
53      3      2.88 [          #|         ]
54      3      3.00 [          #          ]
55      3      3.00 [          #          ]
56      3      3.00 [          #          ]
57      4      3.00 [          #          ]
58      4      3.00 [          #          ]
59      3      3.12 [          #          ]
60      3      3.25 [          |#         ]
61      4      3.38 [          | #         ]
62      2      3.25 [          |#         ]
63      3      3.25 [          |#         ]
64      3      3.25 [          |#         ]
```

Yang count spark: 

--- Yang/Yin dominance runs ---
Positions 02-04: 3 consecutive yin-dominant hexagrams

Odd-vs-even transition parity (McKenna 25/75)

Odd-vs-even transition parity (McKenna's 25/75 observation)

Each transition $d(s_i, s_{i+1})$ is between 1 and 6 line changes.
Theorem 1 says within-pair distances are always even (in $\{2,4,6\}$),
so all 32 within-pair transitions are forced even by C1. The 31
between-pair transitions can be any non-5 value.

McKenna (The Invisible Landscape, 1975) noted that ~25% of King Wen's
transitions are odd-distance and ~75% are even-distance. We report both
the LINEAR reading (the standard 63-transition difference wave) and the
CIRCULAR reading (64 transitions, including the wrap-around $s_{63} \rightarrow s_0$).

Last hexagram (position 64): #42 (101010)
First hexagram (position 1): #63 (111111)
Wrap-around distance: $\text{hamming}(\#42, \#63) = 3$ (odd)

Mode	Total	Odd	%Odd	Even	%Even
LINEAR (63 trans)	63	15	23.8095%	48	76.1905%
CIRCULAR (64 trans)	64	16	25.0000%	48	75.0000%

CIRCULAR parity is EXACTLY 25.0000% / 75.0000% – McKenna's 25/75 holds exactly
when the wrap-around transition is included. Without wrap-around, the linear
split is 23.8095% / 76.1905% – approximately but not exactly 25/75.

--- Structural note ---
All 32 within-pair transitions are forced even by C1 + Theorem 1.
So the parity split is determined entirely by the 31 (or 32 with wrap)
between-pair transitions:
Between-pair transitions: 15 odd, 16 even
Within-pair transitions: 0 odd (forced), 32 even (forced)
Plus wrap-around: 1 odd

--- 'C8' candidate: is the wrap-around parity a structural constraint? ---
The spec's C1-C7 do not constrain the wrap-around $s_{63} \rightarrow s_0$. So if
McKenna's exact 25/75 reflects a real constraint on the sequence rather

than a numerical coincidence, it would imply a new constraint:

C8 (weak): hamming(s_63, s_0) is odd (admits {1, 3, 5}; it narrows to {1, 3} only if C2 is ALSO extended to the wrap, which C1-C7 do not)

C8 (strict): hamming(s_63, s_0) = 3 (King Wen's exact value)

CORRECTED 2026-08-01. This block previously called the weak form restrictive, estimating P(odd) from a uniform prior and concluding it eliminated ~60% of completions. That was wrong, and it contradicted this repository's own proven theorem (SPECIFICATION.md, 'Wrap-around parity is odd').

The weak form eliminates NOTHING: odd wrap-around parity is FORCED by C5, not probable under it. Proof: popcount(x^y) = popcount(x) + popcount(y) (mod 2), so summing over the 63 transitions telescopes -- every interior hexagram appears in exactly two transitions and cancels -- leaving

sum of the 63 transition distances = popcount(s_0) + popcount(s_63) (mod 2).

C5 fixes that multiset as {1:2, 2:20, 3:13, 4:19, 6:9}, whose sum is 211, odd.

Hence hamming(s_63, s_0) = popcount(s_63 ^ s_0) is odd for EVERY C5-satisfying sequence. P(odd) = 1, not the 0.40 previously printed here.

So the question is deductive, not empirical, and no Monte Carlo is needed. Only the STRICT form carries information: constraining the wrap to d = 3 filters ~8.2% of canonical orderings (measured at the 560T canonical), not ~80%.

Neither form is in the formal spec: the weak one is a theorem and adds nothing, and the strict one is King Wen's own value, hence data-like by construction.

Hexagram neighborhoods

Hexagram neighborhoods		
For each hexagram, which others are 'nearby' (differ by only 1 or 2 lines)?		
Dense neighborhoods mean a hexagram has many close relatives; sparse ones are more isolated. This also shows whether King Wen places neighbors near each other in the sequence or scatters them.		

01 ☰ Heaven over Heaven	d=1: [09, 10, 13, 14, 43, 44]	(6 neighbors, mean seq dist: 21.2)
02 ☷ Earth over Earth	d=1: [07, 08, 15, 16, 23, 24]	(6 neighbors, mean seq dist: 13.5)
03 ☵ Water over Thunder	d=1: [08, 17, 24, 42, 60, 63]	(6 neighbors, mean seq dist: 32.7)
04 ☶ Mountain over Water	d=1: [07, 18, 23, 41, 59, 64]	(6 neighbors, mean seq dist: 31.3)
05 ☰ Water over Heaven	d=1: [09, 11, 43, 48, 60, 63]	(6 neighbors, mean seq dist: 34.0)
06 ☱ Heaven over Water	d=1: [10, 12, 44, 47, 59, 64]	(6 neighbors, mean seq dist: 33.3)
07 ☷ Earth over Water	d=1: [02, 04, 19, 29, 40, 46]	(6 neighbors, mean seq dist: 19.0)
08 ☵ Water over Earth	d=1: [02, 03, 20, 29, 39, 45]	(6 neighbors, mean seq dist: 18.7)
09 ☰ Wind over Heaven	d=1: [01, 05, 26, 37, 57, 61]	(6 neighbors, mean seq dist: 26.2)
10 ☷ Heaven over Lake	d=1: [01, 06, 25, 38, 58, 61]	(6 neighbors, mean seq dist: 25.8)
11 ☵ Earth over Heaven	d=1: [05, 19, 26, 34, 36, 46]	(6 neighbors, mean seq dist: 18.7)
12 ☱ Heaven over Earth	d=1: [06, 20, 25, 33, 35, 45]	(6 neighbors, mean seq dist: 17.3)
13 ☷ Heaven over Fire	d=1: [01, 25, 30, 33, 37, 49]	(6 neighbors, mean seq dist: 20.2)
14 ☵ Fire over Heaven	d=1: [01, 26, 30, 34, 38, 50]	(6 neighbors, mean seq dist: 20.2)
15 ☷ Earth over Mountain	d=1: [02, 36, 39, 46, 52, 62]	(6 neighbors, mean seq dist: 28.8)
16 ☵ Thunder over Earth	d=1: [02, 35, 40, 45, 51, 62]	(6 neighbors, mean seq dist: 27.8)
17 ☱ Lake over Thunder	d=1: [03, 25, 45, 49, 51, 58]	(6 neighbors, mean seq dist: 26.2)
18 ☶ Mountain over Wind	d=1: [04, 26, 46, 50, 52, 57]	(6 neighbors, mean seq dist: 25.8)
19 ☵ Earth over Lake	d=1: [07, 11, 24, 41, 54, 60]	(6 neighbors, mean seq dist: 20.5)
20 ☱ Wind over Earth	d=1: [08, 12, 23, 42, 53, 59]	(6 neighbors, mean seq dist: 19.5)
21 ☵ Fire over Thunder	d=1: [25, 27, 30, 35, 38, 51]	(6 neighbors, mean seq dist: 13.3)
22 ☱ Mountain over Fire	d=1: [26, 27, 30, 36, 37, 52]	(6 neighbors, mean seq dist: 12.7)
23 ☱ Mountain over Earth	d=1: [02, 04, 20, 27, 35, 52]	(6 neighbors, mean seq dist: 14.7)
24 ☷ Earth over Thunder	d=1: [02, 03, 19, 27, 36, 51]	(6 neighbors, mean seq dist: 15.0)
25 ☷ Heaven over Thunder	d=1: [10, 12, 13, 17, 21, 42]	(6 neighbors, mean seq dist: 11.5)
26 ☱ Mountain over Heaven	d=1: [09, 11, 14, 18, 22, 41]	(6 neighbors, mean seq dist: 11.8)
27 ☱ Mountain over Thunder	d=1: [21, 22, 23, 24, 41, 42]	(6 neighbors, mean seq dist: 7.8)
28 ☱ Lake over Wind	d=1: [31, 32, 43, 44, 47, 48]	(6 neighbors, mean seq dist: 12.8)
29 ☱ Water over Water	d=1: [07, 08, 47, 48, 59, 60]	(6 neighbors, mean seq dist: 23.5)
30 ☱ Fire over Fire	d=1: [13, 14, 21, 22, 55, 56]	(6 neighbors, mean seq dist: 16.8)
31 ☱ Lake over Mountain	d=1: [28, 33, 39, 45, 49, 62]	(6 neighbors, mean seq dist: 12.7)
32 ☱ Thunder over Wind	d=1: [28, 34, 40, 46, 50, 62]	(6 neighbors, mean seq dist: 12.7)
33 ☱ Heaven over Mountain	d=1: [12, 13, 31, 44, 53, 56]	(6 neighbors, mean seq dist: 16.2)
34 ☱ Thunder over Heaven	d=1: [11, 14, 32, 43, 54, 55]	(6 neighbors, mean seq dist: 15.8)
35 ☱ Fire over Earth	d=1: [12, 16, 21, 23, 56, 64]	(6 neighbors, mean seq dist: 19.7)
36 ☱ Earth over Fire	d=1: [11, 15, 22, 24, 55, 63]	(6 neighbors, mean seq dist: 19.7)
37 ☱ Wind over Fire	d=1: [09, 13, 22, 42, 53, 63]	(6 neighbors, mean seq dist: 19.0)

38	☳	Fire over Lake	d=1: [10, 14, 21, 41, 54, 64]	(6 neighbors, mean seq dist: 19.0)
39	☵	Water over Mountain	d=1: [08, 15, 31, 48, 53, 63]	(6 neighbors, mean seq dist: 18.3)
40	☵	Thunder over Water	d=1: [07, 16, 32, 47, 54, 64]	(6 neighbors, mean seq dist: 18.3)
41	☶	Mountain over Lake	d=1: [04, 19, 26, 27, 38, 61]	(6 neighbors, mean seq dist: 18.5)
42	☴	Wind over Thunder	d=1: [03, 20, 25, 27, 37, 61]	(6 neighbors, mean seq dist: 19.5)
43	☵	Lake over Heaven	d=1: [01, 05, 28, 34, 49, 58]	(6 neighbors, mean seq dist: 20.8)
44	☴	Heaven over Wind	d=1: [01, 06, 28, 33, 50, 57]	(6 neighbors, mean seq dist: 21.2)
45	☳	Lake over Earth	d=1: [08, 12, 16, 17, 31, 47]	(6 neighbors, mean seq dist: 23.8)
46	☴	Earth over Wind	d=1: [07, 11, 15, 18, 32, 48]	(6 neighbors, mean seq dist: 24.8)
47	☵	Lake over Water	d=1: [06, 28, 29, 40, 45, 58]	(6 neighbors, mean seq dist: 16.3)
48	☵	Water over Wind	d=1: [05, 28, 29, 39, 46, 57]	(6 neighbors, mean seq dist: 17.0)
49	☳	Lake over Fire	d=1: [13, 17, 31, 43, 55, 63]	(6 neighbors, mean seq dist: 18.7)
50	☴	Fire over Wind	d=1: [14, 18, 32, 44, 56, 64]	(6 neighbors, mean seq dist: 18.7)
51	☴	Thunder over Thunder	d=1: [16, 17, 21, 24, 54, 55]	(6 neighbors, mean seq dist: 22.2)
52	☶	Mountain over Mountain	d=1: [15, 18, 22, 23, 53, 56]	(6 neighbors, mean seq dist: 22.5)
53	☴	Wind over Mountain	d=1: [20, 33, 37, 39, 52, 57]	(6 neighbors, mean seq dist: 14.7)
54	☳	Thunder over Lake	d=1: [19, 34, 38, 40, 51, 58]	(6 neighbors, mean seq dist: 15.3)
55	☴	Thunder over Fire	d=1: [30, 34, 36, 49, 51, 62]	(6 neighbors, mean seq dist: 13.7)
56	☳	Fire over Mountain	d=1: [30, 33, 35, 50, 52, 62]	(6 neighbors, mean seq dist: 14.3)
57	☴	Wind over Wind	d=1: [09, 18, 44, 48, 53, 59]	(6 neighbors, mean seq dist: 19.2)
58	☳	Lake over Lake	d=1: [10, 17, 43, 47, 54, 60]	(6 neighbors, mean seq dist: 20.2)
59	☴	Wind over Water	d=1: [04, 06, 20, 29, 57, 61]	(6 neighbors, mean seq dist: 30.2)
60	☴	Water over Lake	d=1: [03, 05, 19, 29, 58, 61]	(6 neighbors, mean seq dist: 31.2)
61	☴	Wind over Lake	d=1: [09, 10, 41, 42, 59, 60]	(6 neighbors, mean seq dist: 24.2)
62	☴	Thunder over Mountain	d=1: [15, 16, 31, 32, 55, 56]	(6 neighbors, mean seq dist: 27.8)
63	☳	Water over Fire	d=1: [03, 05, 36, 37, 39, 49]	(6 neighbors, mean seq dist: 34.8)
64	☴	Fire over Water	d=1: [04, 06, 35, 38, 40, 50]	(6 neighbors, mean seq dist: 35.2)

```

--- Neighborhood clustering in King Wen sequence ---
Mean sequence distance between Hamming-1 neighbors: 20.6
Mean sequence distance between any two hexagrams: 21.7
Hamming-1 neighbors are closer than average in the sequence (clustered).

```

```

--- Neighborhood clustering null model ---
Shuffling binary hexagrams 10,000 times to build a null distribution
of mean sequence distance between Hamming-1 neighbors.
Null distribution: min=18.0, mean=21.7, max=25.4
King Wen observed mean: 20.6
Percentile: 11.8% (proportion of shuffles with mean <= King Wen's)
King Wen's neighborhood clustering is within the range expected by chance.

```

Recurrence plot

```

---
Recurrence plot
A recurrence plot shows where the difference wave repeats itself. Each cell (i,j)
is marked if the difference at position i equals the difference at position j.
Diagonal lines indicate repeated sequences; vertical/horizontal lines indicate
values that persist. Clusters of marks reveal recurring local patterns.

```

	1	2	3	4	5	6
	1234567890	1234567890	1234567890	1234567890	1234567890	1234567890
1	+	..	..	..	..	..
2	..	+	..	..	..	..
3	..	..	+	..	..	..
4	..	..	..	+	..	..
5	..	..	..	..	+	..
6	..	..	..	..	..	+
7	..	..	..	..	..	..
8	..	..	..	..	..	..
9	..	..	..	..	..	..
10	..	..	..	..	..	..
11	..	..	..	..	..	..
12	..	..	..	..	..	..
13	..	..	..	..	..	..
14	..	..	..	..	..	..
15	..	..	..	..	..	..
16	..	..	..	..	..	..
17	..	..	..	..	..	..
18	..	..	..	..	..	..
19	..	..	..	..	..	..

20#.....#+.....#.#.....#...#.#.....#.#...#.
21 .#...#.#.##.##.....+##..#.#.#...#...#.#...#.....##...#...
22 .#...#.#.##.##.....#+#..#.#.#...#...#.#...#.....##...#...
23 .#...#.#.##.##.....##+..#.#.#...#...#.#...#.....##...#...
24#.....#.....+.....#.#.....#.#...#.#...#...#...#...
25 ..###..#.#...#.#.....+.....###..#.#...#...###.....#.....
26 .#...#.#.##.##.....###..#+.#.#...#...#.#...#.....##...#...
27 #.....#.....#.....+.#.....#.....#.....#.....#...#...
28 .#...#.#.##.##.....###..#+.#.#...#...#.#...#.....##...#...
29 #.....#.....#.....#+.....#.....#.....#.....#...#...
30#.....#.#...#.....+.#.....#...#.#...#.....#.#...#...
31 .#...#.#.##.##.....###..#.#..+.....#.#...#.....##...#...
32#.....#.#...#.....#+.....#...#.#...#.....#.#...#...
33 ..###..#.#...#.#.....#.....+##..#.#...#...###.....#.....
34 ..###..#.#...#.#.....#.....#+#..#.#...#...###.....#.....
35 ..###..#.#...#.#.....#.....##+..#.#...#...###.....#.....
36 .#...#.#.##.##.....###..#.#..#.....+.....#.#...#...###.....
37 ..###..#.#...#.#.....#.....###..#+.#.#...#...###.....#.....
38 #.....#.....#.....#.#.....+.....#.....#.....#...#...
39 ..###..#.#...#.#.....#.....###..#+.#.#...#...###.....#.....
40#.....#.#...#.....#.#.....+.....#.#...#.....#.#...#...
41 .#...#.#.##.##.....###..#.#.#...#.....+.#...#.....##...#...
42 ..###..#.#...#.#.....#.....###..#.#..+.....#...###.....#.....
43 .#...#.#.##.##.....###..#.#.#...#.....#+.....#.....##...#...
44#.....#.#...#.....#.#.....#.....+.#.#...#...#...#...
45 ..###..#.#...#.#.....#.....###..#.#...#+.....###.....#.....
46#.....#.#...#.....#.#.....#...#...+.#.....#...#...#...
47 .#...#.#.##.##.....###..#.#.#...#.....#...#+.....##...#...
48#.....#.#...#.....#.#.....#...#...+.....#...#...#...
49 ..###..#.#...#.#.....#.....###..#.#...#...#+##.....#.....
50 ..###..#.#...#.#.....#.....###..#.#...#...#+#.....#.....
51 ..###..#.#...#.#.....#.....###..#.#...#...##+.....#.....
52#.....#.....#.....+.....#...
53 #.....#.....#.....#.#.....#.....#.....+.....#...#...
54 .#...#.#.##.##.....###..#.#.#...#.....#.#...+##...#...
55 .#...#.#.##.##.....###..#.#.#...#.....#.#...#+#...#...
56#.....#.#...#.....#.#.....#...#...+.#...#...
57 ..###..#.#...#.#.....#.....###..#.#...#...###.....+.....
58#.....#.#...#.....#.#.....#...#...+.....#...#...
59 .#...#.#.##.##.....###..#.#.#...#.....#...#...##...+.....
60#.....#.....#.....+...
61 #.....#.....#.....#.#.....#.....#.....+.#...
62#.....#.#...#.....#.#.....#...#...#.....#.#...+...
63 #.....#.....#.....#.#.....#.....#.....#.....#...#...+

Recurrence rate: 952/3906 (24.4%)

--- Recurrence rate null model ---
Theoretical expected recurrence rate (sum of p_i^2): 25.6%

King Wen recurrence rate: 24.4%
Mean random recurrence rate: 23.2%
King Wen percentile vs random: 72.4%

DNA codon mapping

DNA codon mapping
Both the I Ching (64 hexagrams) and the genetic code (64 codons) are systems of 64 elements built from binary-like pairs: yin/yang for hexagrams, and the base pairs A-T and G-C for DNA. Each codon is a sequence of 3 bases from the set {A, C, G, U}, giving $4^3 = 64$ possibilities. While this is a numerical coincidence (both are $2^6 = 64$), comparing their structural properties is instructive about combinatorial systems in general.
Caveat: the mapping of bit pairs to DNA bases (00=U, 01=C, 10=A, 11=G) is one of 24 possible assignments. Different mappings produce different codon assignments and different degeneracy statistics. No mapping is privileged by either system. This analysis is illustrative, not evidence of a biological connection.

Pos	Hex	Binary	Codon	Amino Acid	Name
----	-----	-----	-----	-----	----

01	☰	111111	GGG	Gly	Heaven over Heaven
02	☷	000000	UUU	Phe	Earth over Earth
03	☳	010001	CUC	Leu	Water over Thunder
04	☴	100010	AUA	Ile	Mountain over Water
05	☵	010111	CCG	Pro	Water over Heaven
06	☶	111010	GAA	Glu	Heaven over Water
07	☱	000010	UUA	Leu	Earth over Water
08	☲	010000	CUU	Leu	Water over Earth
09	☳	110111	GCG	Ala	Wind over Heaven
10	☴	111011	GAG	Glu	Heaven over Lake
11	☵	000111	UCG	Ser	Earth over Heaven
12	☶	111000	GAU	Asp	Heaven over Earth
13	☷	111101	GGC	Gly	Heaven over Fire
14	☰	101111	AGG	Arg	Fire over Heaven
15	☱	000100	UCU	Ser	Earth over Mountain
16	☲	001000	UAU	Tyr	Thunder over Earth
17	☳	011001	CAC	His	Lake over Thunder
18	☴	100110	ACA	Thr	Mountain over Wind
19	☵	000011	UUG	Leu	Earth over Lake
20	☶	110000	GUU	Val	Wind over Earth
21	☷	101001	AAC	Asn	Fire over Thunder
22	☰	100101	ACC	Thr	Mountain over Fire
23	☱	100000	AUU	Ile	Mountain over Earth
24	☲	000001	UUC	Phe	Earth over Thunder
25	☳	111001	GAC	Asp	Heaven over Thunder
26	☴	100111	ACG	Thr	Mountain over Heaven
27	☵	100001	AUC	Ile	Mountain over Thunder
28	☶	011110	CGA	Arg	Lake over Wind
29	☷	010010	CUA	Leu	Water over Water
30	☰	101101	AGC	Ser	Fire over Fire
31	☱	011100	CGU	Arg	Lake over Mountain
32	☲	001110	UGA	Stop	Thunder over Wind
33	☳	111100	GGU	Gly	Heaven over Mountain
34	☴	001111	UGG	Trp	Thunder over Heaven
35	☵	101000	AAU	Asn	Fire over Earth
36	☶	000101	UCC	Ser	Earth over Fire
37	☷	110101	GCC	Ala	Wind over Fire
38	☰	101011	AAG	Lys	Fire over Lake
39	☱	010100	CCU	Pro	Water over Mountain
40	☲	001010	UAA	Stop	Thunder over Water
41	☳	100011	AUG	Met	Mountain over Lake
42	☴	110001	GUC	Val	Wind over Thunder
43	☵	011111	CGG	Arg	Lake over Heaven
44	☶	111110	GGA	Gly	Heaven over Wind
45	☷	011000	CAU	His	Lake over Earth
46	☰	000110	UCA	Ser	Earth over Wind
47	☱	011010	CAA	Gln	Lake over Water
48	☲	010110	CCA	Pro	Water over Wind
49	☳	011101	CGC	Arg	Lake over Fire
50	☴	101110	AGA	Arg	Fire over Wind
51	☵	001001	UAC	Tyr	Thunder over Thunder
52	☶	100100	ACU	Thr	Mountain over Mountain
53	☷	110100	GCU	Ala	Wind over Mountain
54	☰	001011	UAG	Stop	Thunder over Lake
55	☱	001101	UGC	Cys	Thunder over Fire
56	☲	101100	AGU	Ser	Fire over Mountain
57	☳	110110	GCA	Ala	Wind over Wind
58	☴	011011	CAG	Gln	Lake over Lake
59	☵	110010	GUA	Val	Wind over Water
60	☶	010011	CUG	Leu	Water over Lake
61	☷	110011	GUG	Val	Wind over Lake
62	☰	001100	UGU	Cys	Thunder over Mountain
63	☱	010101	CCC	Pro	Water over Fire
64	☲	101010	AAA	Lys	Fire over Water

Unique amino acids in King Wen order: 21

Total possible amino acids + stop: 21

--- Degeneracy comparison ---

In DNA, many single-base changes preserve the amino acid (the code is 'degenerate').

Do single-line hexagram changes preserve the mapped amino acid?

Single-line changes that preserve amino acid: 100/384 (26.0%)

Shannon entropy analysis

```
---
Shannon entropy analysis
Entropy measures disorder: high entropy means the difference values are spread
evenly across all possibilities (random-looking), low entropy means certain values
dominate (structured). We compare the King Wen wave's entropy against thousands
of random permutations of the same 64 hexagrams. If King Wen's entropy is
unusually low, the sequence is more structured than random chance would produce.
---
King Wen difference wave entropy: 2.0759 bits
Maximum entropy (all 7 values): 2.8074 bits
Maximum entropy (5 observed values): 2.3219 bits
Mean entropy of random permutations: 2.1938 bits
Min random entropy observed: 1.6648 bits
Max random entropy observed: 2.5009 bits
King Wen percentile: 12.4% (lower = more structured)
Effect size (Cohen's d): -1.15 (negative = more structured than random)

--- Entropy conditioned on pair constraint ---
The unconstrained comparison above may be misleading: the pair structure
itself constrains the entropy. How does King Wen compare against random
orderings that also satisfy the pair constraint?
Mean pair-constrained entropy: 2.2336 bits
King Wen percentile (pair-constrained): 5.9%
(Similar to unconstrained percentile of 12.4%.)

--- Distribution comparison ---
Value      King Wen Expected (random avg)
  0           0             0.0
  1           2             6.1
  2          20            15.0
  3          13            20.0
  4          19            15.0
  5           0             5.9
  6           9             1.0
```

Path analysis (graph theory)

```
---
Path analysis (graph theory)
Imagine the 64 hexagrams as cities on a map, where the 'distance' between any
two is the number of lines that differ. The King Wen sequence is a route that
visits all 64 cities. Is it a short, efficient route (nearby hexagrams placed
next to each other), or a long, wandering one? We compare its total distance
against random routes and a greedy 'always go to the nearest unvisited city'
algorithm. This reveals whether the King Wen ordering was optimized for smooth
transitions or had other priorities.
---
King Wen total path length:  211 (sum of all line changes)
Greedy nearest-neighbor:    75
Mean random path length:   192.1
Min random observed:       160
Max random observed:       232
King Wen percentile:       97.7% (lower = shorter path)
Effect size (Cohen's d):    +2.03

--- Theoretical bounds ---
Minimum possible (63 transitions of 1): 63
Maximum possible (63 transitions of 6): 378
King Wen uses 55.8% of maximum path length

--- Pair-constrained comparison ---
The above compares against fully random orderings. A fairer comparison is
against random orderings that also preserve the pair structure (each pair
of hexagrams stays adjacent). This is the right null model for asking
whether King Wen's path length is unusual GIVEN its pair constraint.
Mean pair-constrained path length: 214.2
Min pair-constrained observed:    190
```

Max pair-constrained observed: 239
King Wen percentile (pair-constrained): 28.4%
Effect size (Cohen's d, pair-constrained): -0.49

Constraint satisfaction analysis

```
---
Constraint satisfaction analysis
The King Wen sequence satisfies several constraints simultaneously:
  1. All 32 consecutive pairs are either reverse or inverse (never unrelated)
  2. No consecutive hexagrams differ by exactly 5 lines
  3. No consecutive hexagrams are identical (0-line transitions)
How rare is it for a random ordering to satisfy all these at once? We test
random permutations against each constraint individually and combined.
---
Results from 10,000 random permutations:
All pairs reverse/inverse:      0 (0.000%)
No 5-line transitions:         19 (0.19%)
Both constraints together:      0 (0.0000%)
No random permutation satisfied both constraints.
Statistical note: 0/10,000 gives a 95% upper bound of <0.0300%
(i.e., the true rate is less than ~1 in 3,333, but we cannot
distinguish 'impossible' from 'extremely rare' with this sample size).

--- Conditional probability: P(no-5 | pair constraint) ---
The pair structure constrains transitions within pairs (always even or 6),
so 5-line transitions can only occur at the 31 between-pair boundaries.
How often do pair-constrained orderings also avoid 5-line transitions?
Pair-constrained trials: 100,000
Also satisfy no-5:          4,506 (4.51%)
Approximately 1 in 22 pair-constrained orderings avoid 5-line transitions.
The no-5 property is uncommon but not extraordinary among pair-constrained orderings.

--- Sensitivity analysis: reversed bit convention ---
What if bit 0 = top line instead of bottom? All binary values reverse,
changing pair types and the difference wave. Key properties tested:
Pair structure preserved:  Yes
No-5 property preserved:   Yes
Difference wave identical: Yes
Original entropy: 2.0759, reversed: 2.0759
Original path length: 211, reversed: 211
All key properties are invariant under bit reversal.
```

Bootstrap confidence intervals

```
---
Bootstrap confidence intervals
The Monte Carlo results (e.g., '0.18% of random orderings avoid 5-line
transitions') are estimates based on sampling. Bootstrap resampling quantifies
how precise those estimates are by repeatedly resampling from the results and
computing confidence intervals. Narrower intervals = more reliable estimates.
---
Base trials: 10,000
Bootstrap resamples: 1000

No-5-line-transition rate: 0.140%
95% confidence interval: [0.070%, 0.220%]
Interval width: 0.150 percentage points

Approximately 1 in 714 random orderings
95% CI: 1 in 455 to 1 in 1429

Note: These CIs measure the precision of the Monte Carlo estimate, not
fundamental uncertainty about the true proportion. They reflect how much
the estimate would vary if you re-ran the simulation with 10,000 trials.
Increasing --trials narrows these CIs because the estimate becomes more precise.
```

Monte Carlo analysis

Monte Carlo analysis (10,000 random permutations)

The King Wen sequence has a striking property: no two consecutive hexagrams differ by exactly 5 lines. With 6 lines per hexagram and 7 possible difference values (0-6), is avoiding 5 remarkable or just a coincidence? To find out, we randomly shuffle the 64 hexagrams thousands of times and check how often a random ordering also avoids 5-line transitions. The rarer it is, the less plausible chance becomes as an explanation for the avoidance.

Permutations with no 5-line transitions: 11/10,000 (0.11%)

Approximately 1 in 909 random orderings share this property.

Odds ratio against random: 908:1